KON 313
Feedback Control Systems

Block Diagrams

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Contents

Introduction

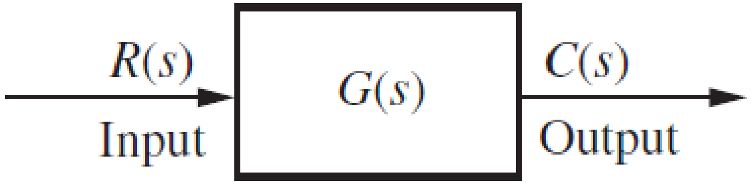
- A **block diagram** of a system is a pictorial representation of the functions performed by each component and of the flow of signals.
- Differing from a purely abstract mathematical representation, a block diagram has the advantage of indicating more realistically the signal flows of the actual system.
- Systems are generally composed of multiple subsystems. In a block diagram all subsystems are linked to each other through functional blocks.

Components Used in Block Diagrams

Signals. Note that the signal can pass only in the direction of the arrows.

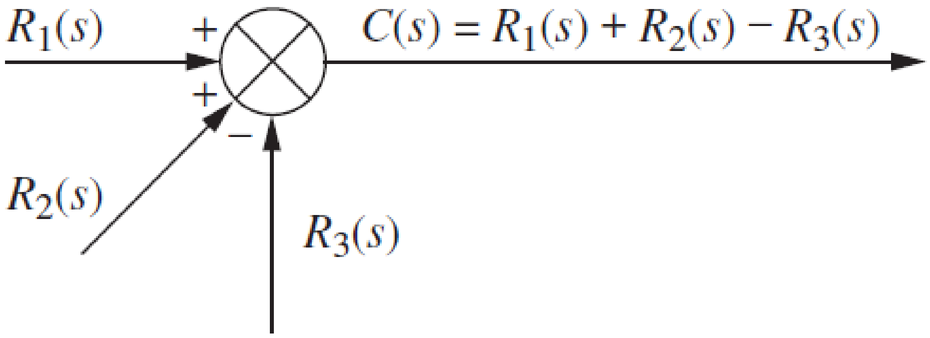
Signals

Blok. A subsystem is represented as a block with an input, an output, and a transfer function.



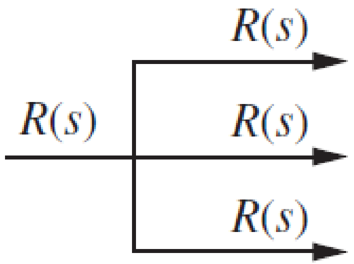
System

Summing junction. The output signal is the algebraic sum of the input signals. Any number of the inputs can be present.



Summing junction

Pickoff point/ Branch point. Distributes the input signal.



Pickoff point

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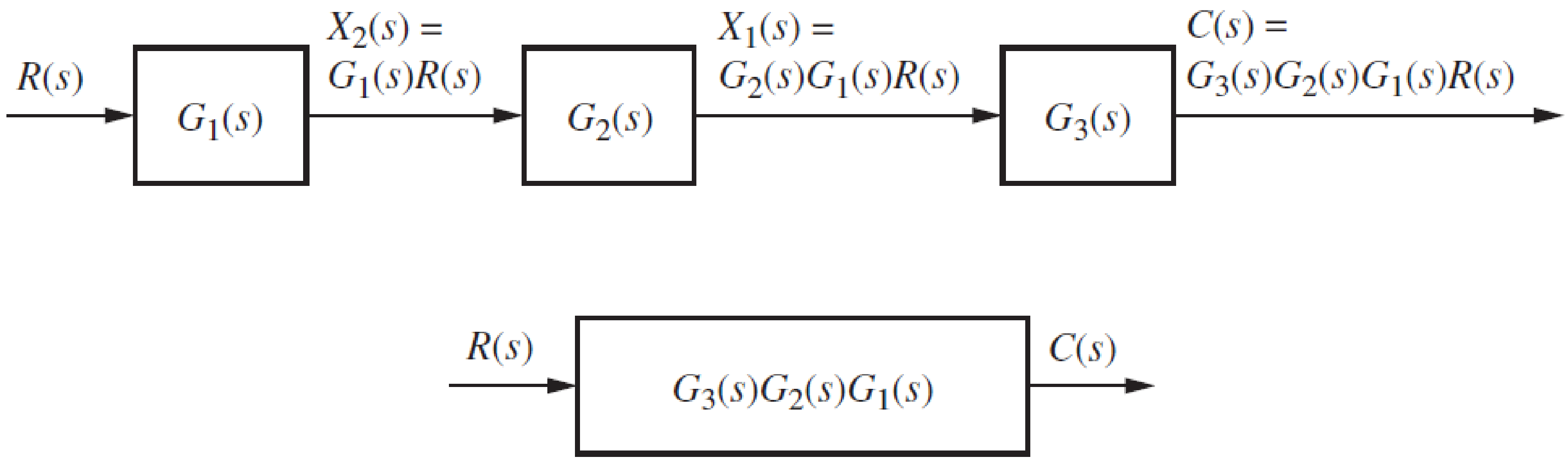
Block Diagrams Algebra

- Since the response of a single transfer function can be calculated, we want to connect and then represent multiple subsystems as a single transfer function.
- We will develop techniques to reduce each representation to a single transfer function. **Block diagram algebra** will be used to reduce block diagrams
- There are some common topologies for interconnecting subsystems in order to derive the single transfer function representation.
  1. Cascade Connection
  2. Parallel Connection
  3. Feedback Connection

Cascade Connection

The equivalent transfer function, $G_e(s)$, is the product of the subsystems' transfer functions.



$$G_e(s) = G_3(s)G_2(s)G_1(s)$$

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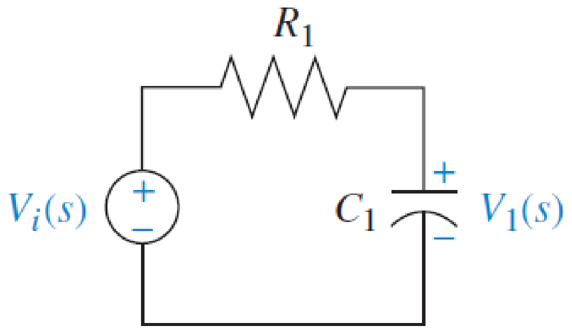
Loading Issue in Cascade Connection

Loading Issue. The cascaded blocks equation is derived under the assumption that interconnected subsystems do not load adjacent subsystems. i.e. A subsystem's output should remain the same whether the subsequent subsystem is connected or not.

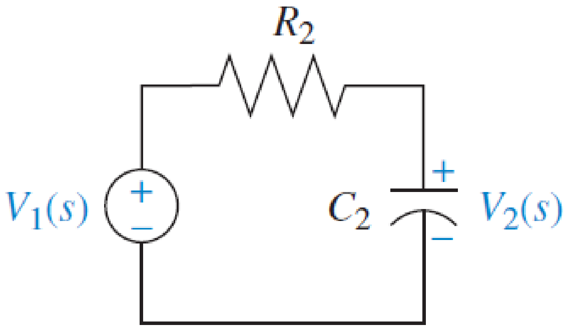
One way to prevent loading is to use an amplifier between the two networks. The amplifier has a **high-impedance input**, so that it does not load the previous network. It has a **low-impedance output**, so that it looks like a pure voltage source to the subsequent network.

Loading Issue in Cascade Connection

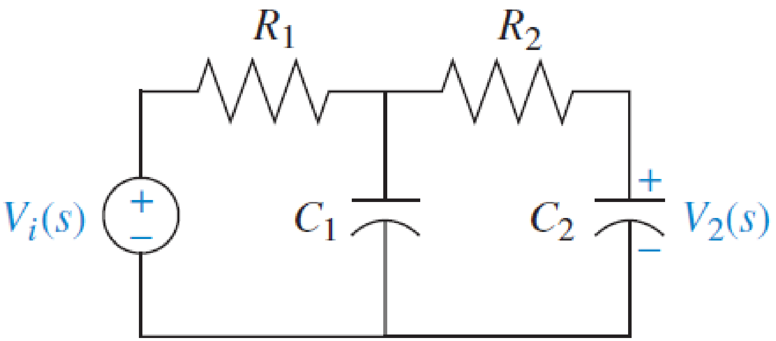
E.g.



$$G_1(s) = \frac{V_1(s)}{V_i(s)} = \frac{\frac{1}{R_1C_1}}{s + \frac{1}{R_1C_1}}$$

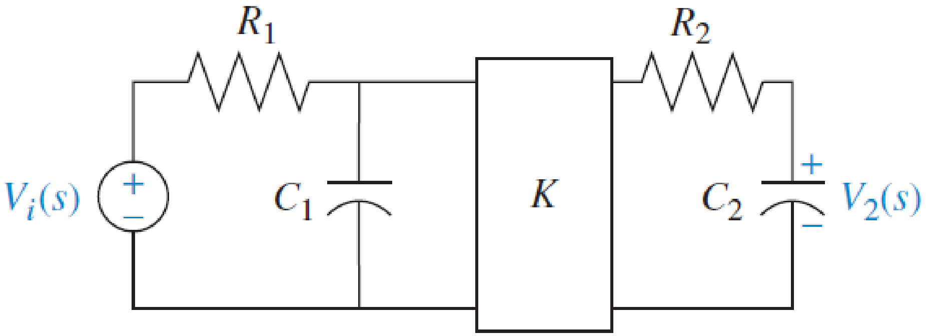


$$G_2(s) = \frac{V_2(s)}{V_1(s)} = \frac{\frac{1}{R_2C_2}}{s + \frac{1}{R_2C_2}}$$



$$G_T(s) = \frac{V_2(s)}{V_i(s)} \neq G_2(s)G_1(s)$$

$$G(s) = \frac{V_2(s)}{V_i(s)} = \frac{\frac{1}{R_1C_1R_2C_2}}{s^2 + \left(\frac{1}{R_1C_1} + \frac{1}{R_2C_2} + \frac{1}{R_2C_1}\right)s + \frac{1}{R_1C_1R_2C_2}}$$

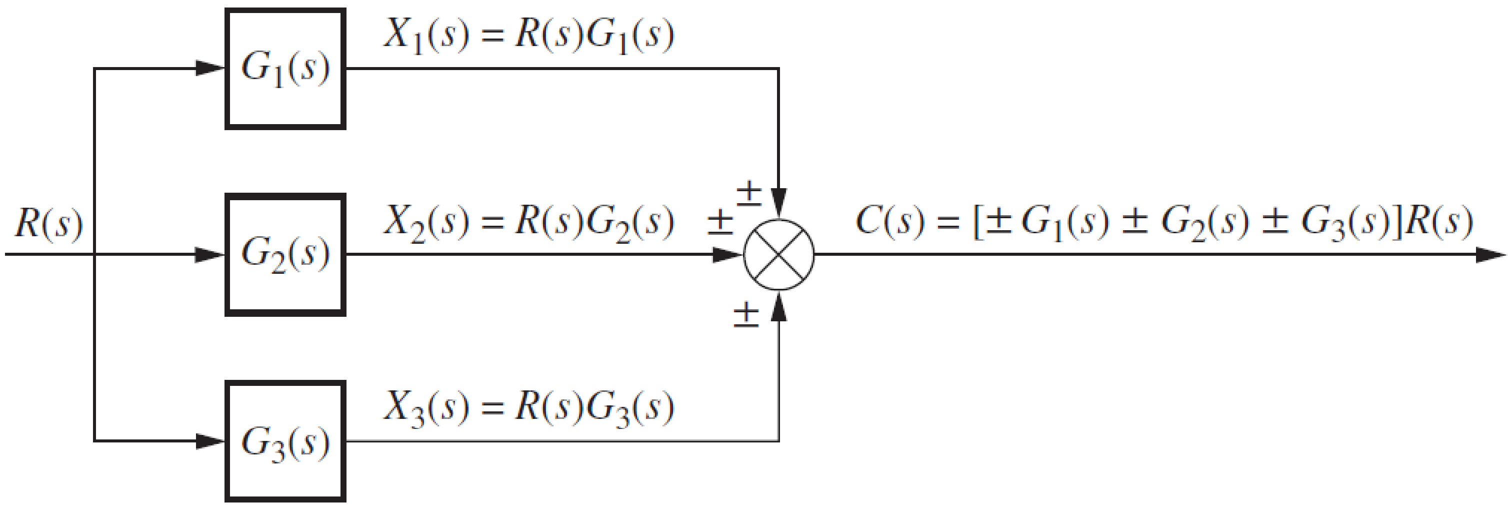


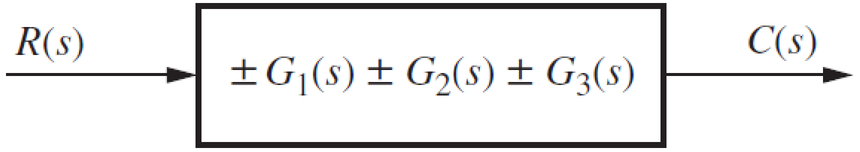
$$G_T(s) = \frac{V_2(s)}{V_i(s)} = KG_2(s)G_1(s)$$

$$G_2(s)G_1(s) = \frac{\frac{1}{R_1C_1R_2C_2}}{s^2 + \left(\frac{1}{R_1C_1} + \frac{1}{R_2C_2}\right)s + \frac{1}{R_1C_1R_2C_2}}$$

Parallel Connection

Parallel subsystems have a common input and an output formed by the algebraic sum of the outputs from all the subsystems.





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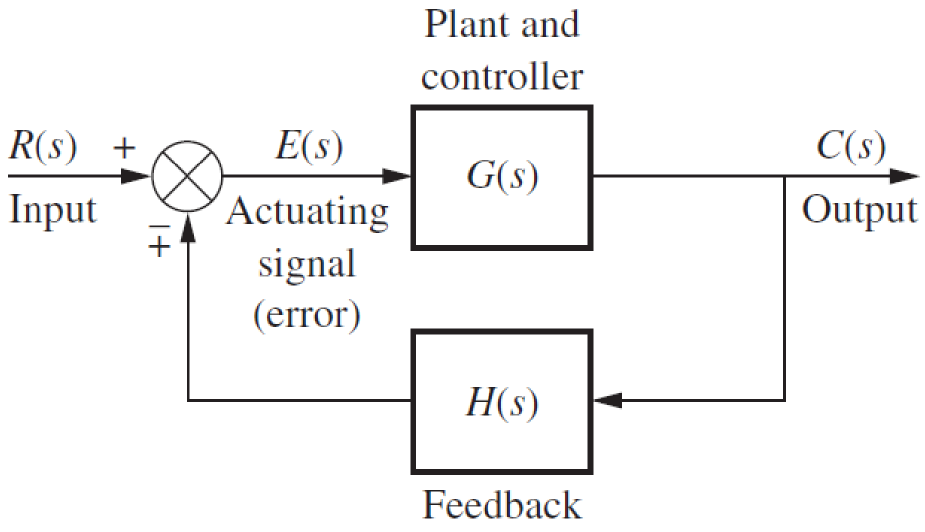
Feedback Connection

The feedback system forms the basis for our study of control systems engineering. Remind the advantage of closed-loop, or feedback control, systems over open-loop systems.

$$E(s) = R(s) \mp C(s)H(s)$$

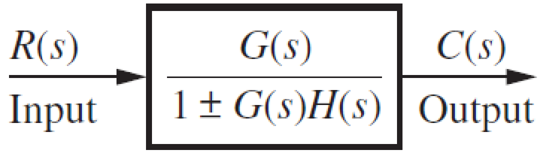
$$C(s) = E(s)G(s) \quad \Rightarrow \quad E(s) = \frac{C(s)}{G(s)}$$



by substituting $E(s)$, **closed loop tf** is:

$$G_e(s) = \frac{G(s)}{1 \pm G(s)H(s)}$$

The product, $G(s)H(s)$, is called the **open-loop transfer function**, or loop gain.



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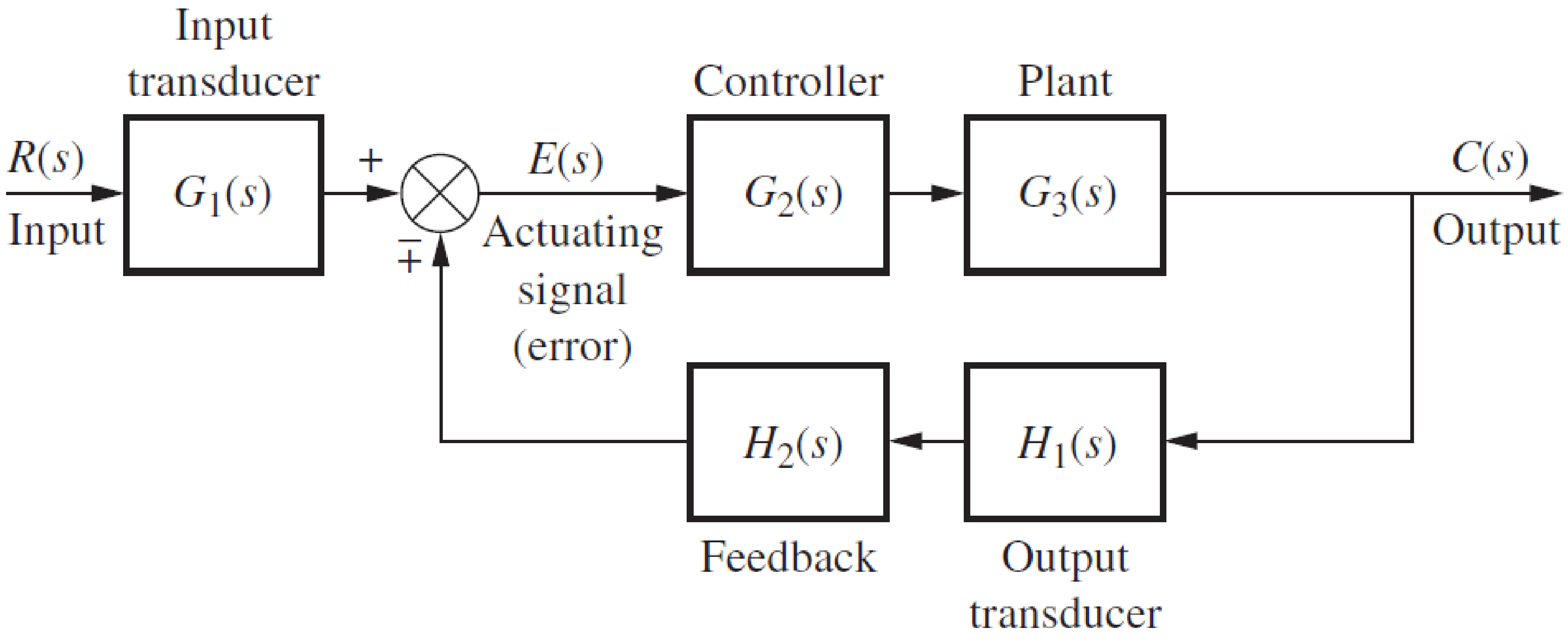
Feedback Connection

The more complete model:



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Moving Blocks to Create Familiar Forms

- Note that the familiar forms (cascade, parallel, and feedback) are not always apparent in a block diagram.

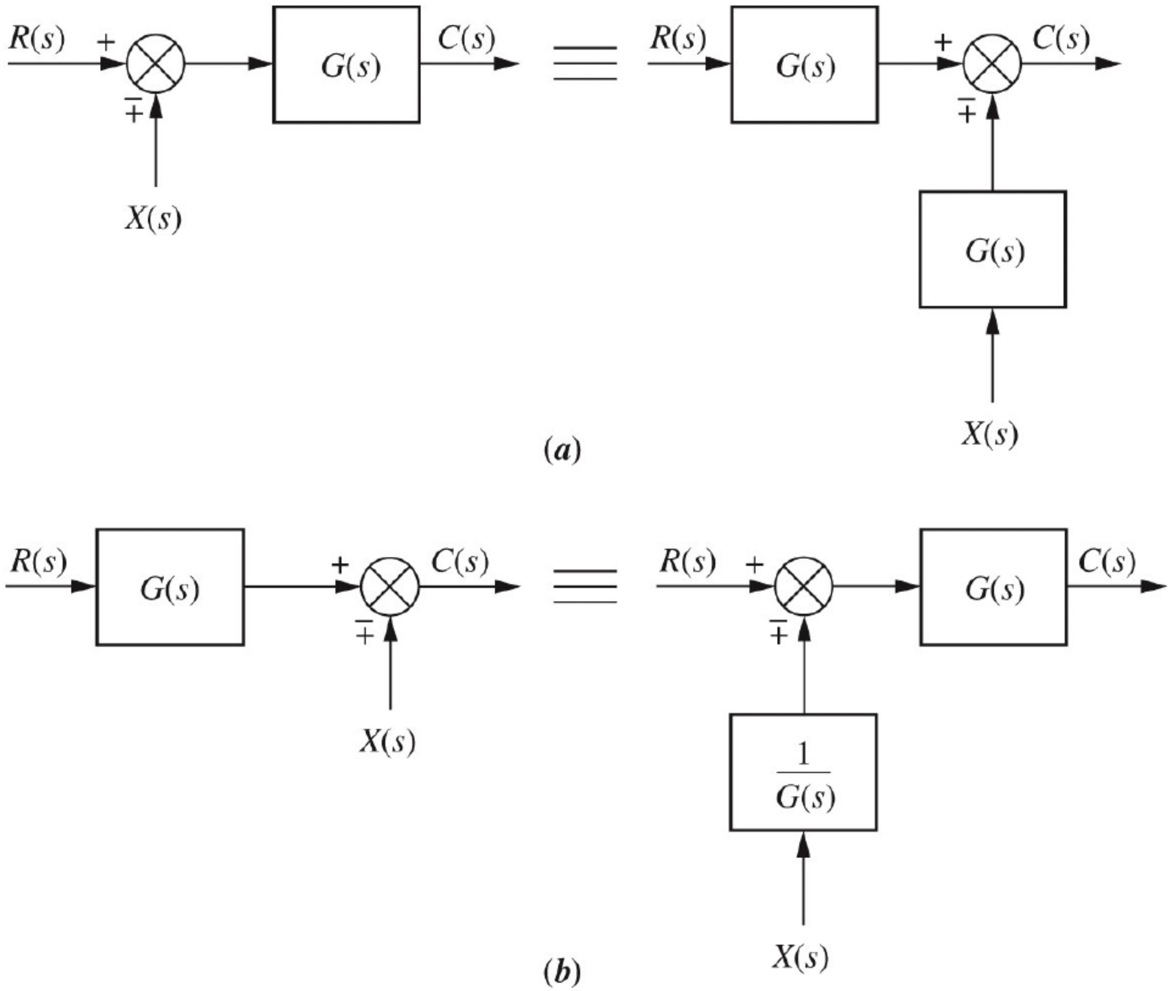
- We sometimes need to make block moves in order to establish familiar forms when they almost exist.

  1. Moving a block left or right past a summing junction

  2. Moving transfer functions left or right past a pickoff point

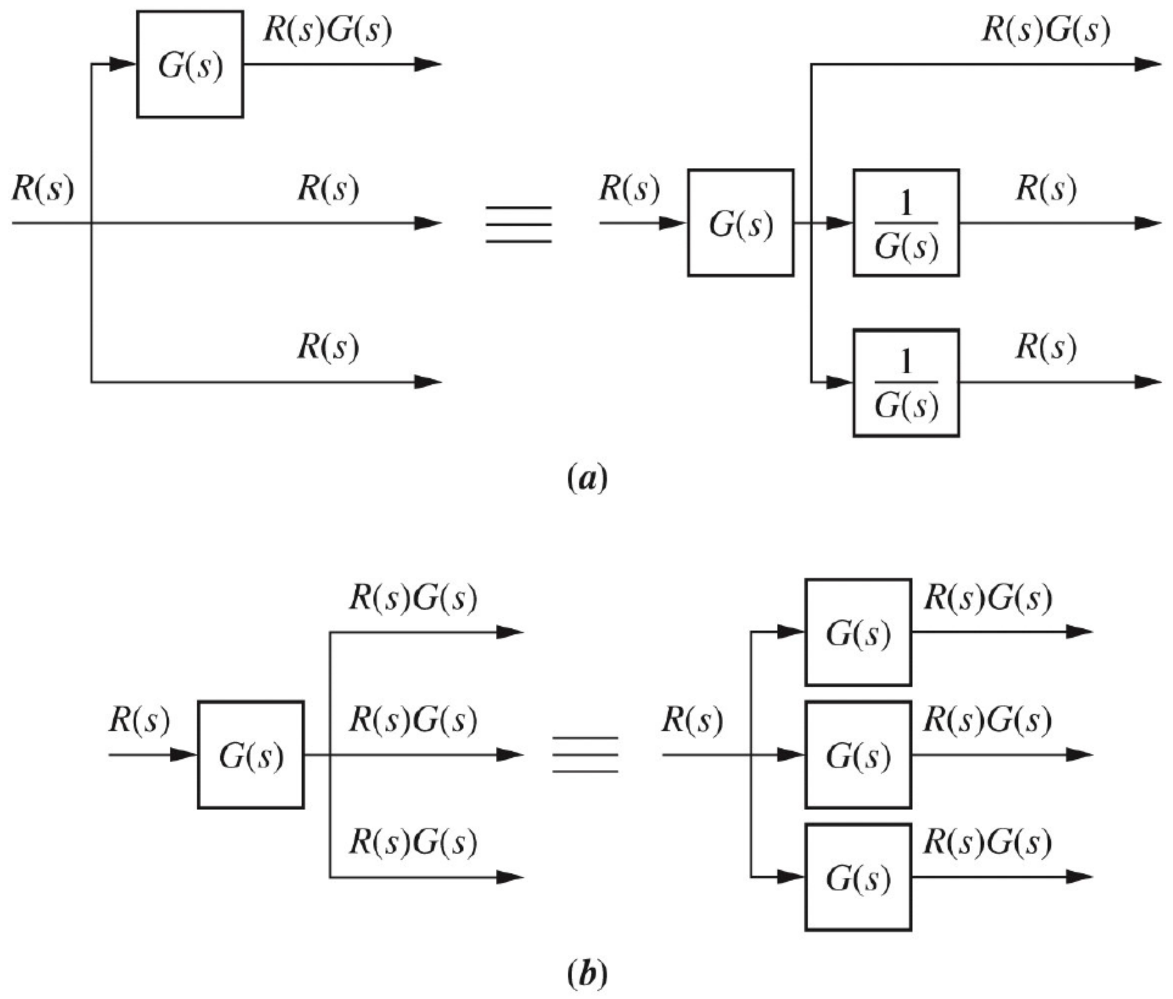
Moving Blocks to Create Familiar Forms

1. Moving a block left or right past a summing junction



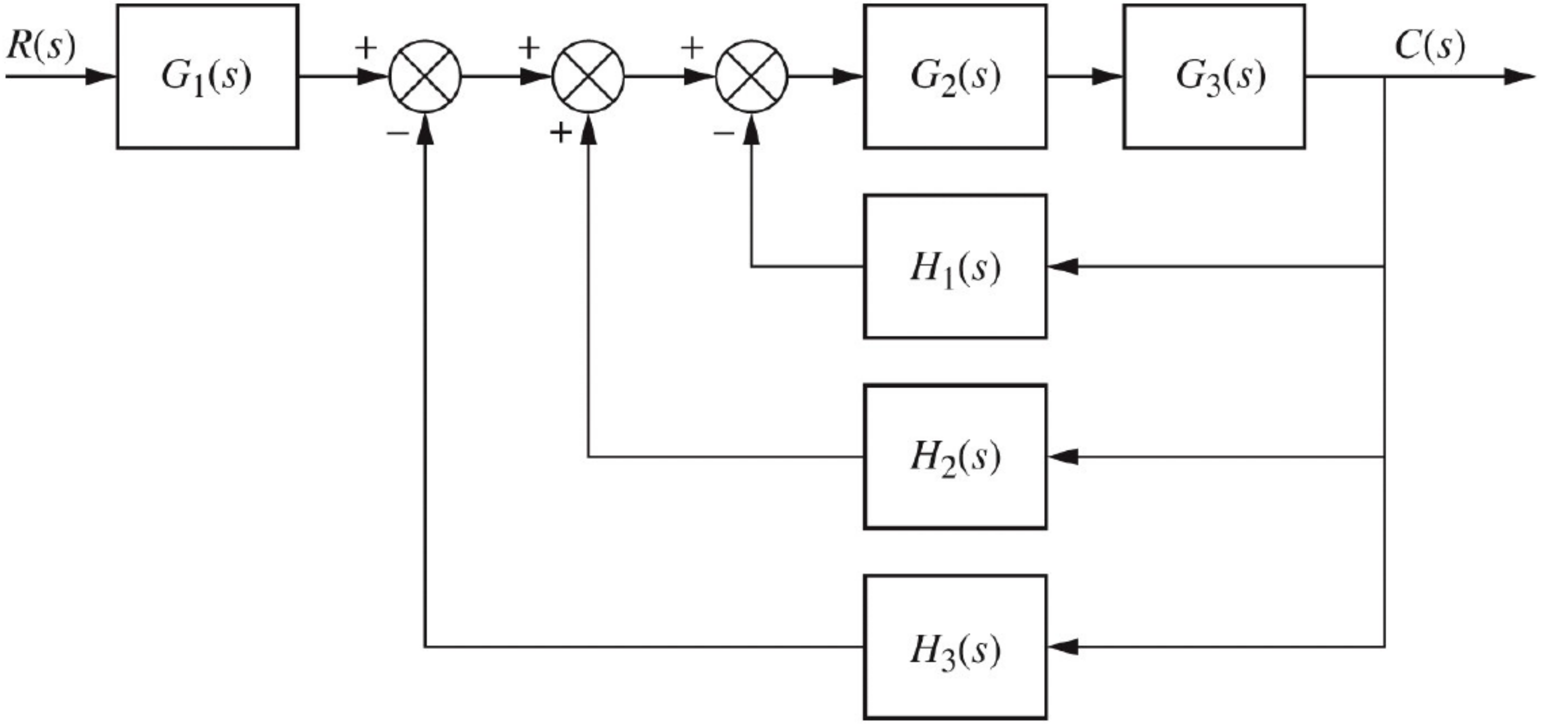
Moving Blocks to Create Familiar Forms

2. Moving transfer functions left or right past a pickoff point



Block Diagrams Reduction, Some Examples

Example 1



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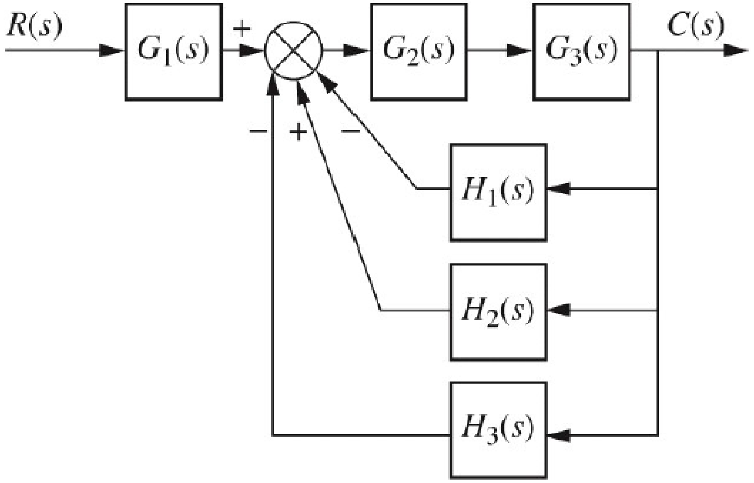
Block Diagrams Reduction, Some Examples

Example 1 - Solution

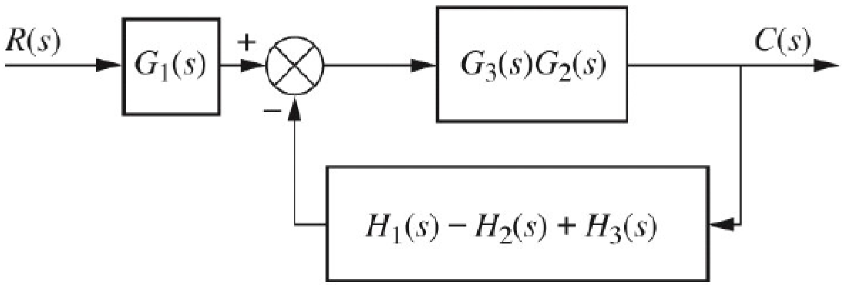
a. Turning the three summing junctions into a single summing junction

b. A parallel connection and a cascade connection

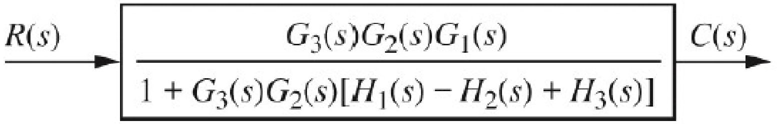
c. Feedback connection, cascaded by $G_1(s)$



(a)



(b)



(c)

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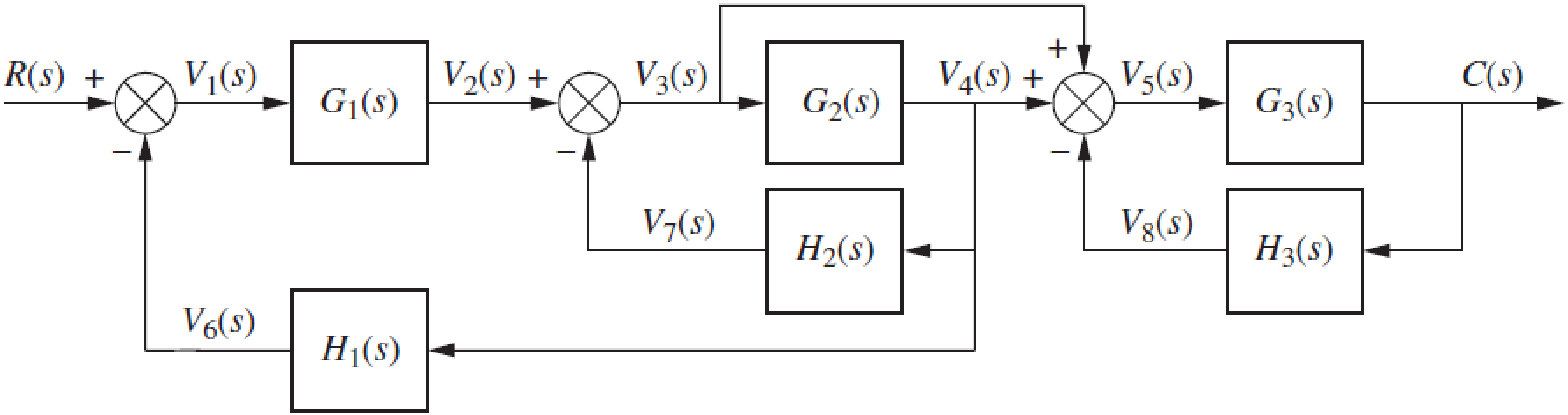
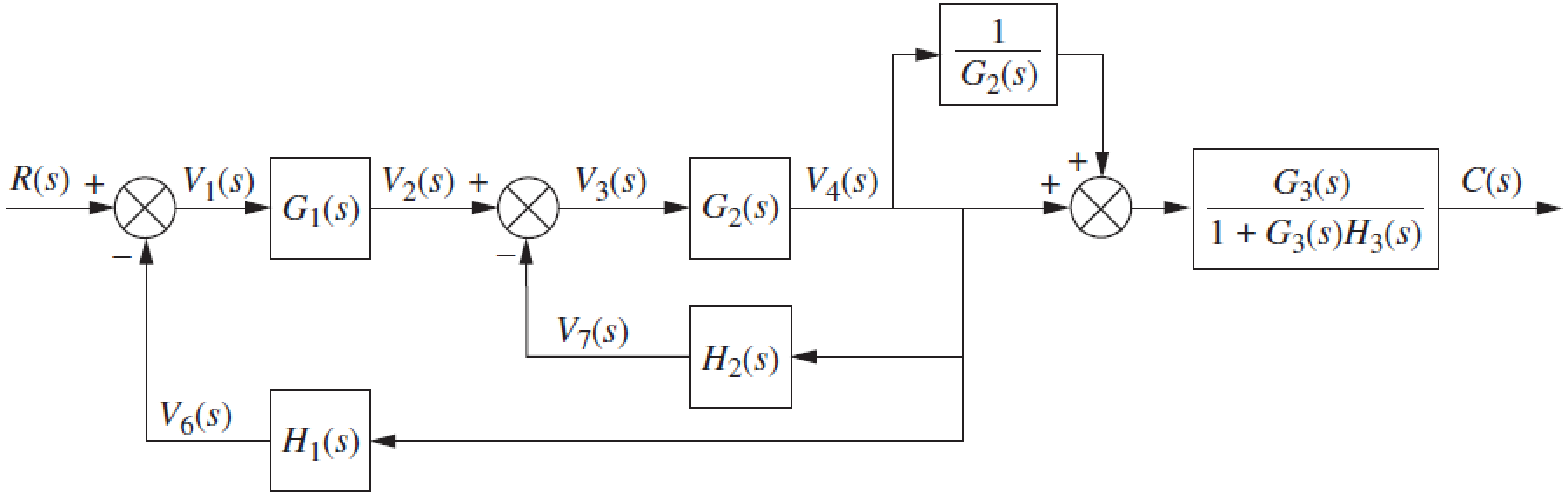
Block Diagrams Reduction, Some Examples

Example 2

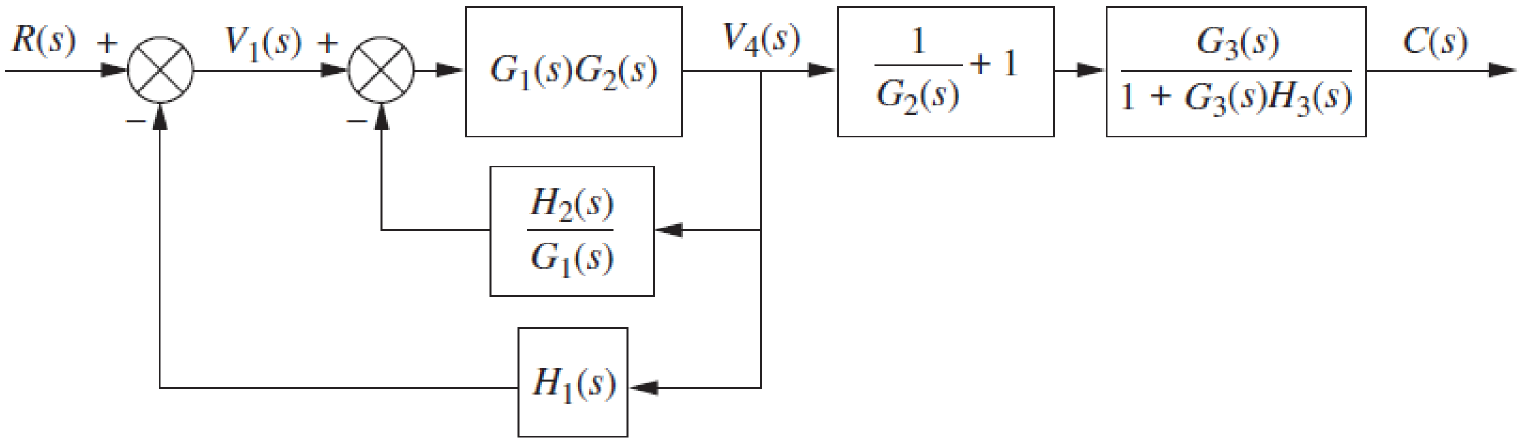


Block Diagrams Reduction, Some Examples

Example 2 - Solution



(a)



(b)

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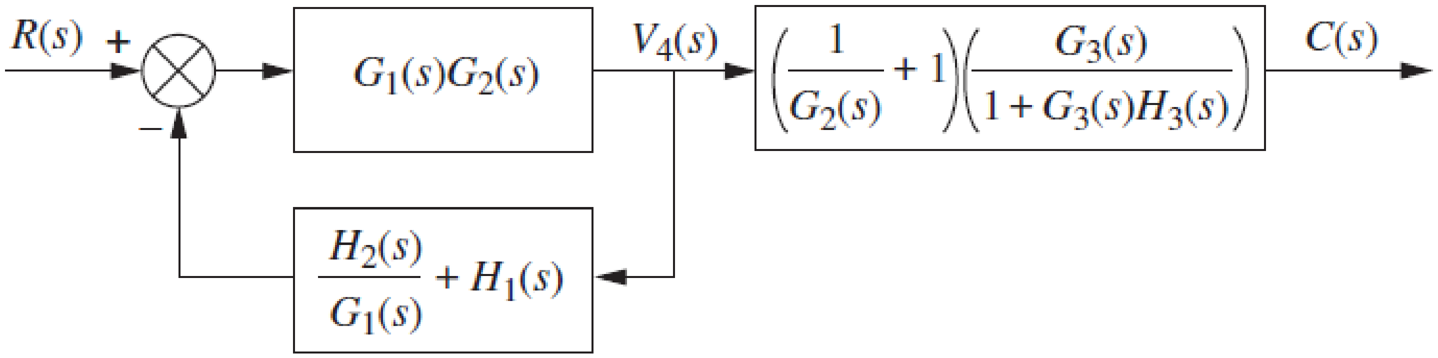
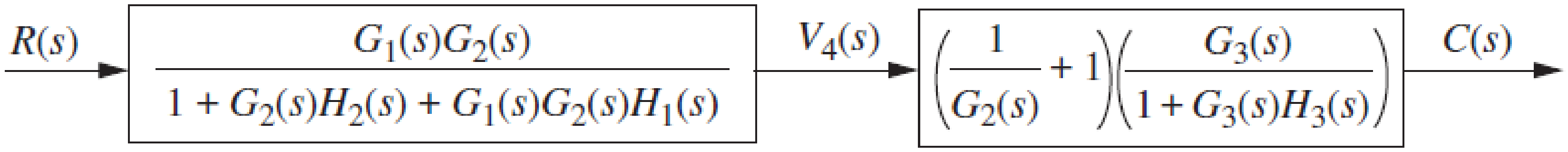
Block Diagrams Reduction, Some Examples

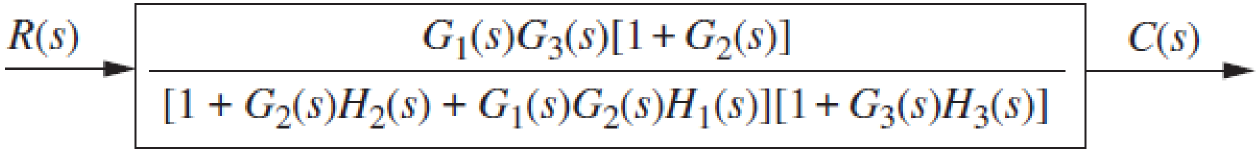
Example 2 – Solution cont.



(*c*)



(*d*)



(*e*)

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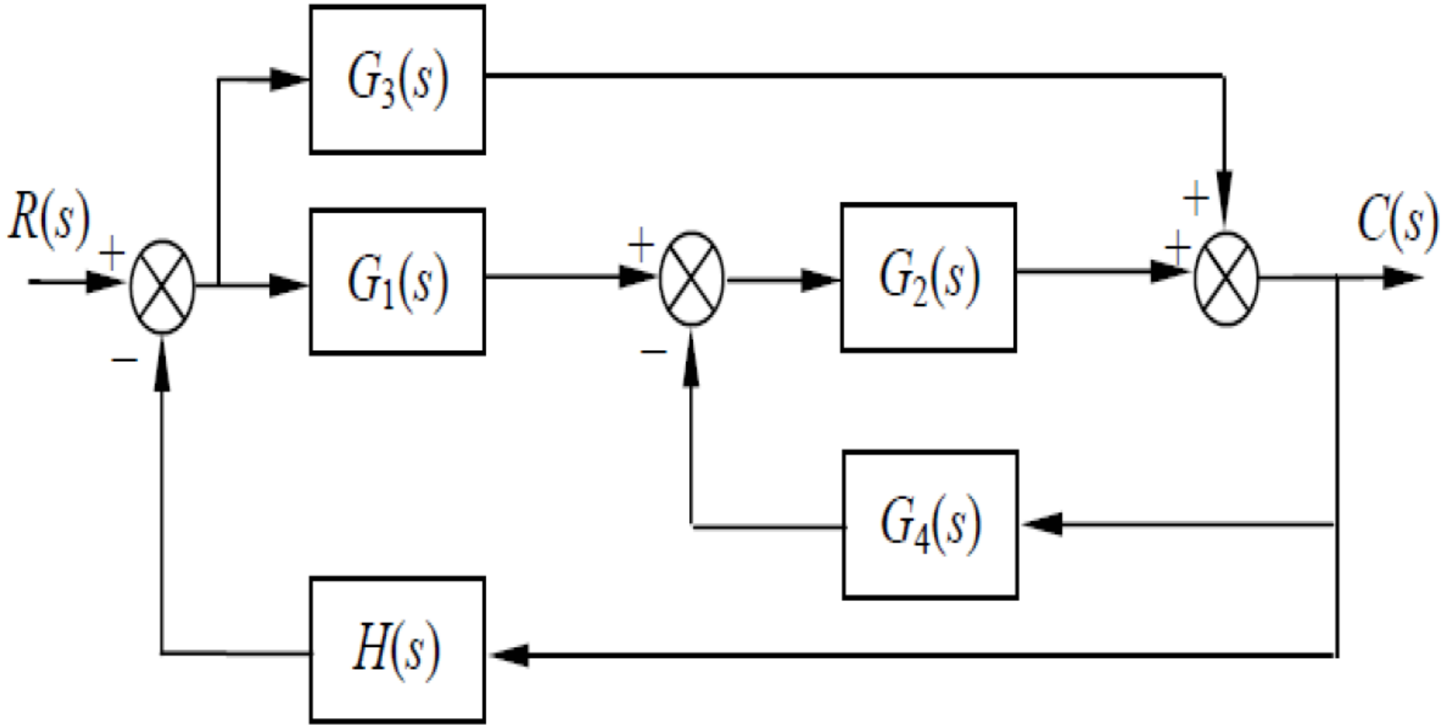
Block Diagrams Reduction, Some Examples

Example 3



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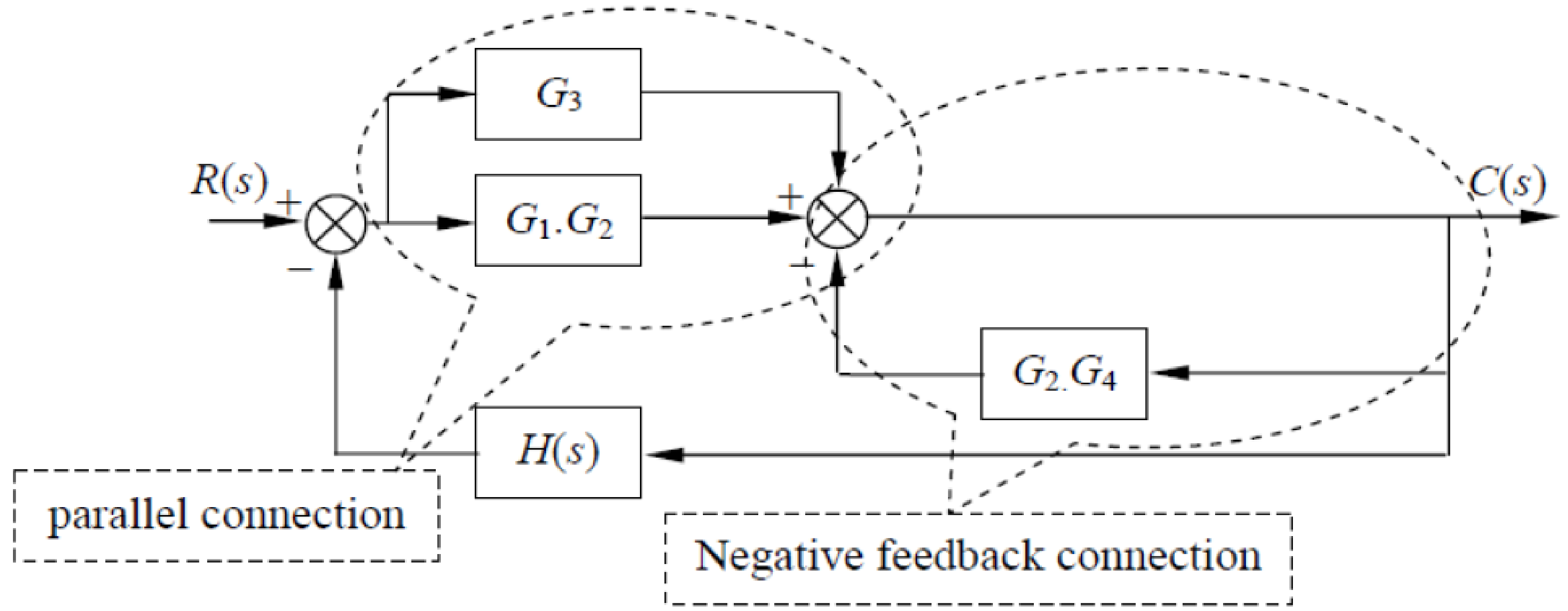
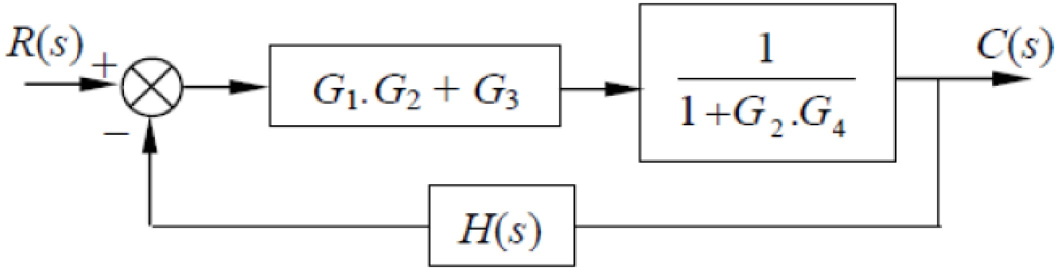
Block Diagrams Reduction, Some Examples

Example 3 – Solution 1

Move $G_2(s)$ to the left past the summing junction to combine two summing junctions:





$$\frac{C(s)}{R(s)} = \frac{G_1(s).G_2(s) + G_3(s)}{1 + G_2(s).G_4(s) + H(s).[G_1(s).G_2(s) + G_3(s)]}$$

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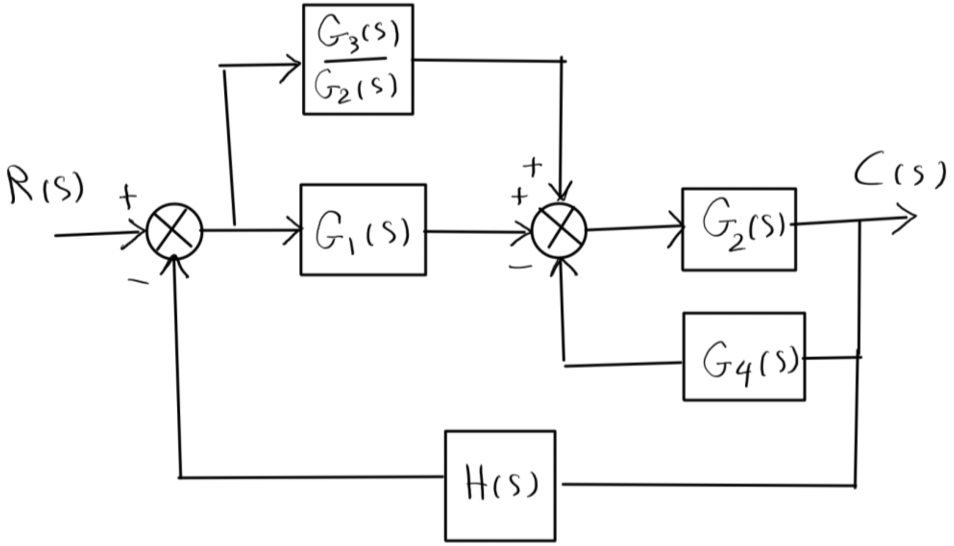
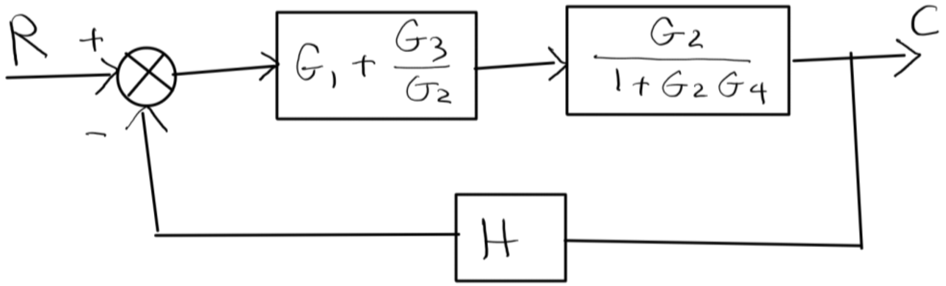
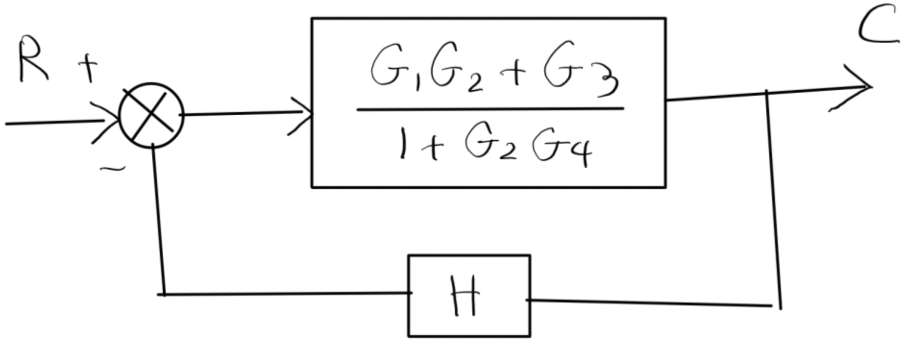
Block Diagrams Reduction, Some Examples

Example 3 – Solution 2







$$G_e = \frac{\dfrac{G_1G_2+G_3}{1+G_2G_4}}{1+\dfrac{H[G_1G_2+G_3]}{1+G_2G_4}}$$

$$= \frac{G_1G_2+G_3}{1+G_2G_4+H[G_1G_2+G_3]}$$

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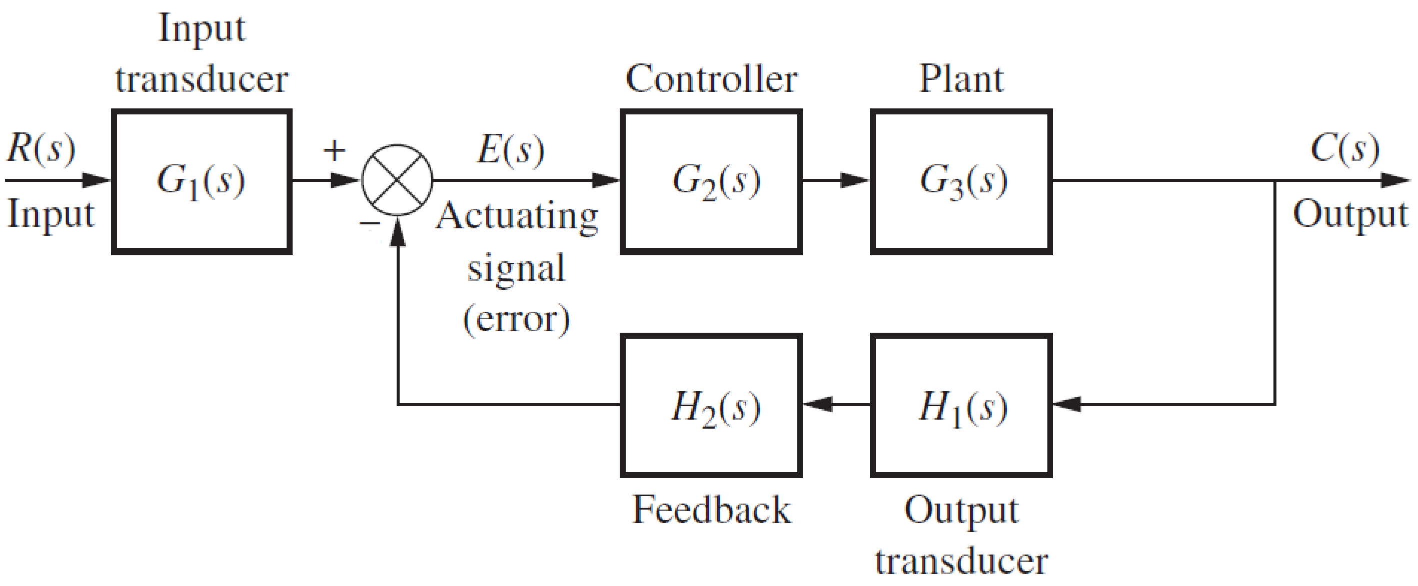
Q4-1

Find the transfer function of the following system.



a) $\dfrac{G_1G_2G_3}{1+H_1H_2}$

b) $\dfrac{G_1G_2G_3}{1-H_1H_2}$

c) $\dfrac{G_1G_2G_3}{1+H_1H_2G_2G_3}$

d) $\dfrac{G_1G_2G_3}{1-H_1H_2G_2G_3}$

e) None of the above

Answer: c

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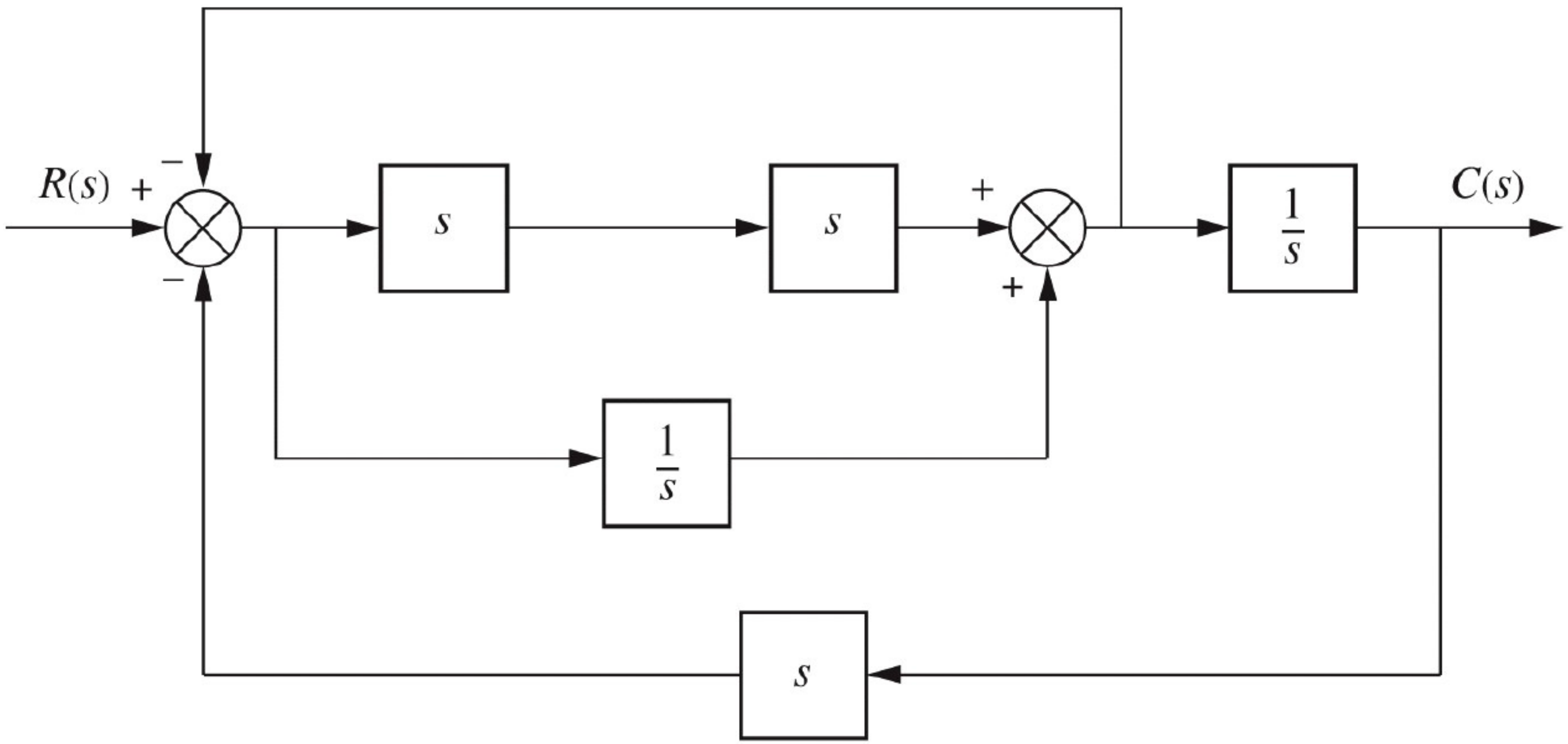
Q4-2



Answer: $T(s) = \dfrac{s^3 + 1}{2s^4 + s^2 + 2s}.$

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